

Remote Charging Methods and Apparatus

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ABSTRACT

The present invention is related to remote charging methods and apparatus to achieve remote charging using lasers and optical rectifiers.

BACKGROUND OF THE INVENTION

1. FIELD OF THE INVENTION

The present invention relates to methods and apparatuses of remote charging. More particularly, this invention relates to remote charging with lasers based on optical rectification in second order nonlinear optical materials.

2. DESCRIPTION OF THE RELATED ART

In the modern world, consumers are enticed by cordless, quick charging, and long battery-life per charge devices. This demand is evident when the success of wireless headphones and cellphones is considered. Thus, it has become more and more important for technology to adapt and become longer-lasting, quicker charging, and remote chargeable. However, these

three attributes are rarely seen together within a single power source (e.g. battery). The cause of this is due to the relation between the charging time and discharging time within a standard re-chargeable lithium-ion battery: as the charge time decreases, the discharge time also decreases. This effect makes a long lasting and extremely quick charging device almost impossible. This disadvantage of current power sources greatly limit the applications and convenience of devices and can be considered as the bottleneck in the energy storage industry. The demand for a solution to this bottleneck can further be demonstrated in remote sensing applications. In this field, many sensors are deployed to an area where accessibility by road transportation would be impossible. Therefore, according to those conditions, it would be extremely difficult (if not impossible) to maintain the sensors over a long period of time. Therefore, in these applications, remote charging would be the only solution to solve this accessibility problem. Not only can remote charging solve this issue of a limited battery lifetime, but it will also remove the need for a large power source/harvester such as a solar panel or a radio frequency (RF) energy harvesting antenna. Another example of the demand for a remotely re-chargeable power source is seen in satellites and space exploration/research. Within the vast expanse of space, satellites and probes must travel large distances to conduct their missions. In some cases, these places include a very dark area where sunlight sometimes cannot reach the solar panels of the satellite. In these instances, a remotely chargeable device which can charge quickly and maintain its charge over long periods of time would be ideal. In addition, since solar panels contribute a portion of satellites and probes' weight, removing the need for solar panels to give payload to other devices is another proponent to the remote charging method. Thus, from the above examples, it is evident that there is a very strong demand for a remotely rechargeable power source that can charge quickly and discharge slowly.

There are already several existing energy storage and battery technologies. Some of such methods are already commercially available. The most popular and regularly used type of battery in today's world is the lithium-ion battery. The lithium-ion battery uses chemical compounds to store chemical energy. As shown in **Fig.1**, the chemical energy can then be converted to electrical energy when a lithium-ion battery **1** is connected to a circuit load **2** through metal wires **3**. This is done by using a cathode material such as lithium cobalt oxide and an anode material such as lithium-carbon (or graphite) to create a potential difference between the cathode and anode. Through the chemical reaction processes of oxidation and reduction, chemical energy is converted to electric energy. In this process, electrons are driven by the electric field **E**, formed by the potential difference between the two electrodes, and cycled through the load circuit to form an electric current. In fact, the lithium-ion battery has been disclosed in many literatures, such as U.S. Patent No. 9130231 B2, September 8, 2015. Ion C. Halalay, et al., Lithium-Ion Battery; U.S. Patent No. 5939864 A, August 17, 1999. Stephen J. Lenhart, et al., Lithium-Ion Battery Charge Control Method.

Another type of energy storage and battery technology is the supercapacitor battery. In contrast to the normal lithium-ion battery that stores energy chemically, supercapacitor batteries stores energy electrostatically. Supercapacitors achieve this by using a dielectric material **1** which is placed between two metal electrode plates **2a** and **2b**, as shown in **Fig.2**. When an electric field is applied across the two electrodes, opposite charges are formed on the two electrodes. This process is called the charging process for the supercapacitor battery. Since the dielectric material is an insulator, even after the electric field is removed, the positive and negative charges can be held in place, maintaining the potential difference between the two electrodes. When a charged supercapacitor battery is connected to an electric load through metal wires **3**, an electron flow (i.e. electric current) can be generated, as shown

in **Fig.2**. This process is called the discharging process of the supercapacitor battery. As a result of its electrostatic nature, supercapacitors allow for extremely quick charging and discharging. Another key feature of a supercapacitor is its capacitance, in other words, its ability to store charge. To increase the capacitance of a normal capacitor, the distance between the electrode plates is decreased and surface area of the electrode plates is enlarged significantly, by using a folded or roll structure. A supercapacitor differs from a regular capacitor in its ability to store charge (i.e. energy). The supercapacitor battery can significantly increase the ability to store charge without increasing the volume. In other words, the capacitance of a superconductor is tremendously higher than a regular capacitor. In fact, the battery technique using supercapacitors has been disclosed in several literatures, such as US Patent No. 8094433 B2, January 10, 2012. Zhao-Wu Tian, et al., Supercapacitor; US Patent No. 6762926 B1, July 13, 2004. Lih-Ren Shiue, et al., Supercapacitor with High Energy Density.

One method of remote charging is based on the widely known and very popular photovoltaic technology. Through a light source, such as the Sun, photovoltaic cells absorb photons from the light source to create electron-hole pairs. As shown in **Fig.3**, when the electrons in a solar cell **1** with a PN junction structure absorb photon energy from the light source **4**, electron-hole pairs are generated in the depletion region of the solar cell. Due to the electric field established in the depletion region from N-type layer to P-type layer, electrons with negative charge move toward the N-type layer, creating potential difference between the electrodes on P-type and N-type layers. This effect, also known as the photovoltaic effect, creates a potential difference between the P-type and N-type semiconductors which can drive electron flow (i.e. electric current) within an electric circuit when the solar cell is connected to an electric load **2** through metal wires **3**. In this sense, the photovoltaic solar cell converts light energy into electrical energy and can be considered as a kind of “remote charging”. Photovoltaic cells have been disclosed in many literatures, such as US Patent No. 20070204899 A1, September 6, 2007. Darry Lafayette Boyd S.R, Photovoltaic Cell: A Solar Amplification Device; US Patent No. 8609984 B2, December 17, 2013. Indranil Bhattacharya, et al., High Efficiency Photovoltaic Cell for Solar Energy Harvesting.

Energy transfer through inductive coupling is another method of remote charging. It is another form of wireless charging. Inductive charging is based on the Faraday's principle of electromagnetic induction which states that whenever a conductor is placed in a varying magnetic field, an electromotive force is induced upon the conductive material. In this remote charging method, electromagnetic induction is used to charge a battery. To achieve this, a time varying magnetic field is generated within the first coil of a charging device, and a second coil (connected to a battery) is placed within the first coil's magnetic field to help supply electric current to the battery. If the conductor circuit is closed, a current is induced, which is called induced current. Through this principle of electromagnetic induction, energy can be transferred to different devices remotely. Remote charging technology based on the inductive coupling has been disclosed, such as US Patent No. 8975864 B2, March 10, 2015. Hyunmin Kim, Wireless Charging System for An Electric Vehicle, and Charging Method for Same (charging through induction).

Yet another method of remote charging is demonstrated through radio frequency (RF) energy harvesting technology. To achieve remote charging with RF waves, RF energy is converted to electric energy. With a RF antenna, an impedance matching circuit, and a diode rectifier, radio waves can be captured and converted to DC electric current. Therefore, when a RF transmitter is directed towards the RF energy harvesting device, remote charging can be

achieved. Many examples of RF energy harvesting devices have been disclosed, such as US Patent No. 20170187228 A1, June 29, 2017. Alister HOSSEINI, et al., Near Field Transmitters for Wireless Power Charging; US Patent No. 8621245 B2, December 31, 2013. John G. Shearer, et al., Powering Devices Using RF Energy Harvesting; US Patent No. 20160181873 A1, June 23, 2016. Paul David Mitcheson, et al., RF Energy Harvester; US Patent No. 8,621,245 B2, John G. Shearer, et al., Powering Devices Using RF Energy Harvesting; US Patent No. 8,975,864 B2, Mar. 10, 2015, Hyunmin Kim, Wireless Charging System for An Electric Vehicle, and Charging Method for Same; Le-Giang Tran, et al., Micro and Nano Syst Lett (2017) 5:14, RF power harvesting: a review on designing methodologies and applications.

Furthermore, remote charging methods using plasmonic devices has also been proposed. In this method, a metal-insulator-metal (MIM) device is used to convert light energy of infrared (IR) and visible wavelengths to electric current. When light is illuminated on the MIM device, surface plasmons are excited which are bounded to the metal-insulator interfaces, generating “hot electrons”. If the number of the “hot electrons” is different between the two metal-insulator interfaces of the MIM device, a potential difference is created. When a charged MIM device is connected to an electric load, a DC current can be created to flow through the electric load. Therefore, a plasmonic device can be considered as a remotely chargeable energy storage device, as it does not require any external power source to charge. Remote charging technology based on the plasmonic devices has been disclosed, such as US Patent No. 2007/0240757 A1, Zhifeng Ren, et al., Solar Cells Using Arrays of Optical rectennas; Matthew T. Sheldon, et al., Science, Oct. 30, 2014, Vol. 346, Issue 6211, pp.828-831, Plasmoelectric potentials in metal nanostructures.

However, while there are many different reported methods of “remote charging” and energy storage, each potential solution has several issues and limitations. Firstly, while lithium-ion batteries have a high energy density, a limited lifetime and an inability of remote charging become major disadvantages. As a result of these limitations listed above, the charging of lithium-ion batteries must rely on electric charging methods. In addition, lithium-ion batteries cannot provide sufficient power and energy to the electric load when the size of the power source is decreased. Just like most batteries, the voltage and current that can be provided by the lithium-ion batteries decreases dramatically after many charge-discharge cycles and have a very limited energy storage capacity when the lithium-ion batteries are miniaturized. Therefore, considering the limitations listed above, lithium-ion batteries cannot be considered as a solution to the bottlenecks of the current energy storage devices and batteries mentioned above (especially remote charging capability).

Second, even though supercapacitors can charge and release the stored energy extremely quick, remote charging a supercapacitor is still impossible at this moment. The charging of a supercapacitor battery must also rely on the electric charging methods. In addition, the capacitance of a supercapacitor, while much higher than a regular capacitor, still cannot compete with other energy storage devices. Furthermore, through the inverse relationship between the capacitance of a supercapacitor and the voltage produced (as the capacitance increases, the voltage decreases at a decreasing rate), it is impossible for a normal supercapacitor to optimize both the capacitance and voltage to a desirable value. Therefore, supercapacitor batteries also cannot overcome the bottlenecks of energy storage devices and batteries mentioned above (especially remote charging capabilities).

Third, even though solar panels are widely accepted into our world as a “power source” capable of harvesting light energy, solar panels are large, bulky, and cannot operate without direct sunlight. Therefore, solar cells are not the solution to the bottlenecks of the current energy storage devices and batteries mentioned above (especially small size).

Fourth, although it has been widely used in wireless cell phone charging, remote charging through inductive coupling has several disadvantages. These drawbacks include a long charging duration and a very restrictive distance between the battery and the charger. Since the force of an electromagnetic field drops quickly with the increase of distance between the battery and the charger, inductive charging is not suitable for long-distance charging. Also, inductive charging requires a pair of coils (which cannot be too small) and a small distance between them to reach maximum efficiency. As a result of the reasons listed above, inductive charging is not the solution to the bottlenecks of the current energy storage devices and batteries mentioned above (especially long-distance remote charging capabilities).

Fifth, although RF energy harvesting technology is getting more and more mature, there are still many drawbacks. First, RF energy extracted from free space usually possesses low power density as the electric field power density decreases quickly with the increase of the distance between RF transmitter and RF receiver. As a result, remote charging through RF energy harvesting has very low efficiency. In addition, RF energy harvesting devices require antennas and RF to DC converters. Therefore, the device is very difficult to minimize and has a very short transmission distance. For these reasons, RF energy harvesting is not the solution to the bottlenecks of the current energy storage devices and batteries mentioned above (especially long-distance remote charging capabilities).

Finally, remote charging based on the surface plasmons also has several issues. Although the concept of using a rectifying antenna (rectenna) to collect solar energy was first proposed in 1972, even after much research and time, efficiency of this remote charging method is still very low. Researchers are still working hard to increase the efficiency by using new nanomaterials such as carbon nanotubes (US Patent No. 2007/0240757 A1, Zhifeng Ren, et al., Solar Cells Using Arrays of Optical rectennas) and metal nanostructures (Matthew T. Sheldon, et al., Science, Oct. 30, 2014, Vol. 346, Issue 6211, pp.828-831, Plasmonic potentials in metal nanostructures). In addition, it is difficult to excite the surface plasmons differently at the two metal-insulator interfaces of the MIM device so that a large potential difference can be created. Therefore, special methods must be used to couple the light into the MIM devices in certain angles and configuration. Furthermore, the number of “hot electrons” generated by the plasmons depends on the area of the metal-insulator interface. Therefore, to enhance energy storage capacity, a large metal-insulator interface is necessary, implying that it is difficult to minimize the MIM device while also maintaining a high energy storage capacity.

3. SUMMARY OF THE INVENTION

The objective of the present invention is to provide methods to overcome the problems involved in the current methods of remote charging by employing a laser source (i.e. converting laser light energy to electric energy) through optical rectification in a second order nonlinear medium. The present invention is designed to provide remote charging methods that are not restricted by the distance of the energy storage device from the light source.

According to one aspect of the present invention, as shown in **Fig.4**, an optical rectifier **1** is connected to a load **2** through metal wires **3**. The optical rectifier **1** is formed by a second

order nonlinear optical material placed between two electrodes. A collimated beam of light from a laser source **4** is focused on the optical rectifier **1** by a focusing optics **5**. The optical rectifier **1** can generate DC electric field within the optical rectifier **1** and create positive and negative charges on the two sides of the optical rectifier **1**, respectively. The separation of positive and negative charges results in potential differences between the two electrodes of the optical rectifier **1**, generating an electric field within the metal wire **3**, and driving electrons to flow through the electric load **2**. This electric current resulting from the optical rectifier **1** generates a DC current following through the load **2**. The laser source **4** can remotely provide energy to the optical rectifier **1** through the focus optics **5** and convert light energy to electric energy through the optical rectifier **1**. The electric load **2** receives electrical energy from the laser source **4** and uses the electrical energy to power the electric load **2** (such as an IC circuit or a re-chargeable battery).

4. BRIEF DESCRIPTION OF THE DRAWINGS

The present invention can be understood more clearly from the detailed description given herein below, in conjunction with the accompanying drawings.

In the drawings:

FIG.1 is a schematic diagram of the first prior art that achieves energy storage using chemical methods (i.e. lithium-ion battery) while providing a DC current to an electric load.

FIG.2 is a schematic diagram of the second prior art that achieves energy storage through electrostatic methods (i.e. supercapacitor battery) while providing a DC current to an electric load.

FIG.3 is a schematic diagram of the third prior art that achieves light energy harvesting using the photovoltaic effect (i.e. solar cell) while providing a DC current to an electric load.

FIG.4 is a structural drawing for explaining the concept of the method described in the first preferred embodiment to achieve remote charging while providing a DC current to an electric load according to the present invention.

FIG.5 is a structural drawing for explaining the details of the optical rectifier to achieve remote charging while providing a DC current to an electric load according to the present invention.

FIG.6 is a simulated plot of the generated voltage across the optical rectifier V_{OR} as a function of the average power of a pulse laser P_{Ave} .

FIG.7 is a simulated plot of the voltage across the optical rectifier V_c as a function of the discharging time.

FIG.8 is a structural drawing for explaining the method described in the second preferred embodiment to achieve remote charging and provide a DC current to an electric load.

5. DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

The present invention solves the foregoing problems by the means described below.

In the first preferred embodiment, the structure of the device is shown in **Fig.4**. As shown in **Fig.4**, an optical rectifier **1** is connected to a load **2** through metal wires **3**. The optical rectifier **1** is formed by a second order nonlinear optical material placed between two electrodes. A beam of light from a laser source **4** is focused on the optical rectifier **1** by a focusing optics **5**. The focus optics **5** can be packaged together with the optical rectifier **1** or set close enough to the optical rectifier **1** so that precise alignment between the laser source **4** and optical rectifier **1** becomes unnecessary. The optical rectifier **1** can generate DC electric field within the optical rectifier **1** and create positive and negative charges on the two sides of the optical rectifier **1**, respectively. The separation of positive and negative charges resulting in a potential difference between the two electrodes of the optical rectifier **1** (as bound charges), which generates an electric field within the metal wire **3**, driving electrons to flow through the electric load **2** when the optical rectifier **1** is connected to the load through the wires **3**. The electric current resulting from the optical rectifier **1** thus generates a DC current following through the load **2**. The laser source **4** with a collimated beam can remotely provide the necessary energy to the optical rectifier **1** to “charge” the power source. Through the focus optics **5** and the optical rectifier **1**, the light energy is converted into electric energy. The electric load **2** receives electrical energy from the laser source **4** and uses the electrical energy to power the electric load **2** (such as an IC circuit or re-chargeable battery).

The operation principle of the optical rectifier **1** can be understood as follows. Electric field $\mathbf{E} = (E_x, E_y, E_z)$ of the electromagnetic (EM) wave from the laser source **4** can be described as:

$$\mathbf{E} = \frac{1}{2} \mathbf{E}_0 (e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} + c.c) = \mathbf{E}_0 \cos(\mathbf{k} \cdot \mathbf{r} - \omega t) \quad (1)$$

Where $\mathbf{E}_0$ is a vector, representing amplitude and direction of the electric field of the EM wave. Direction of $\mathbf{E}_0$ corresponding the polarization direction of the laser beam and light intensity is proportional to the square of $|\mathbf{E}_0|$. $\mathbf{k} = (k_x, k_y, k_z)$ is a wave vector, indicating the direction of light propagation. $\mathbf{r} = (x, y, z)$ presents a position in space. ω is the frequency of the coherent light from the laser source **4** and t is time. When the electric field interacts with an optical nonlinear medium with a second order nonlinear coefficient $\chi^{(2)}$, polarization $\mathbf{P}$ is generated inside the nonlinear medium, given by:

$$\mathbf{P} = \varepsilon_0 \chi^{(1)} \mathbf{E} + \varepsilon_0 \chi^{(2)} : \mathbf{E} \mathbf{E} \quad (2)$$

In general, polarization $\mathbf{P} = (P_x, P_y, P_z)$ is also a vector and $\chi^{(2)}$ is a second rank tensor represented by multiple non-zero elements $\chi_{ijk}^{(2)}$ (where $i, j, k = x, y, z$). From Eqs. (1) and (2), the polarization generated in the nonlinear medium is simply given by:

$$\mathbf{P} = \varepsilon_0 \chi^{(1)} \mathbf{E}_0 \cos(kz - \omega t) + \frac{1}{2} \varepsilon_0 \chi^{(2)} \mathbf{E}_0^2 + \frac{1}{2} \varepsilon_0 \chi^{(2)} \mathbf{E}_0^2 \cos(2kz - 2\omega t) \quad (3)$$

The second term in Eq. (3) is the physical origin of optical rectification, which does not change with time. The optical rectification process generates a bound charge distribution and induces a static electric field across the nonlinear crystal. Therefore, the optical rectification can result in a potential difference between the two surfaces of the nonlinear medium, resulting in an electrical voltage and current if the electrodes are formed properly on the two sides of the nonlinear medium and adequately connected to an electric load. It is worth noting

that this effect of “charging” is uninhibited by the resistance or the capacitance of the electric circuit, and the potential difference is produced almost instantaneously (at the speed of light) when the light source couples into the optical rectifier. Obviously, as shown in Eq.(3), to achieve a large optical rectification, two conditions must be satisfied: (1) a large nonlinear coefficient (i.e. $\chi^{(2)}$) and (2) a large electric field amplitude (i.e. $|E_0|$). $\chi^{(2)}$ is a material property of the nonlinear medium, which is usually dependent on the wavelength and polarization direction of incoming light, and E_0 is dependent on the intensity of light (i.e. light intensity is proportional to square of E_0). Therefore, a coherent light with a narrow linewidth from a laser should be used to achieve large optical rectification.

As shown in Eq. (3), to enhance the ability of generating DC electric field within the optical rectifier, several requirements need to be satisfied. First, materials with high second order optical nonlinear coefficient $\chi^{(2)}$ should be used. Crystals such as dielectric nonlinear optical crystals (lithium niobate LiNbO_3 , potassium titanyl phosphate KTP, etc.), semiconductors (gallium arsenide GaAs, gallium phosphide GaP, etc.), and electro-optical (EO) polymers can be used. Nano-materials and low-dimensional materials (2D and 1D) have also been reported with large $\chi^{(2)}$ nonlinear coefficients (Delwin L. Elder and Larry R. Dalton, *Ind. Eng. Chem. Res.* 2022, 61, 3, 1207–1231, Organic Electro-Optics and Optical Rectification: From Mesoscale to Nanoscale Hybrid Devices and Chip-Scale Integration of Electronics and Photonics). It is worth noting that, since nonlinear coefficient $\chi^{(2)}$ is usually a second rank tensor, light polarization and propagation direction, as well as optical rectifier orientation must be carefully designed so that the largest $\chi^{(2)}$ component can be used. For example, if a EO polymer with a largest $\chi^{(2)}$ component $\chi_{111}^{(2)}$ (i.e. $\chi_{xxx}^{(2)}$) is used, light polarization should be set along x-axis, and light propagation direction can be set along z-axis (Jihua Zhang, et al., *OPTICS EXPRESS* 27504, Vol. 22, No. 22, 3 November 2014, On-chip high-speed optical detection based on an optical rectification scheme in silicon plasmonic platform). Of course, if a polarization insensitive nonlinear medium is used, such as GaAs and GaP, there is no need to align polarization direction of the laser beam with the nonlinear medium orientation since the $\chi^{(2)}$ of this material is independent on light polarization direction (P. S. Kuo, et al., *OPTICS LETTERS*, Vol. 32, No. 18, 2735 – 2737, GaAs optical parametric oscillator with circularly polarized and depolarized pump).

The second requirement which needs to be satisfied is the enhancement of the light intensity within the second order nonlinear medium. This can be achieved by employing a focused beam, and/or a waveguide structure and/or a pulse laser. For example, optical rectification using a large $\chi^{(2)}$ EO polymer with a nanoscale slotted waveguide has already been proposed for ultra-high speed photo-detection (US20070009200A1, Michael Hochberg, et al., Apparatus and method for detecting optical radiation). It has also been demonstrated that, by means of the nanoscale slotted waveguide geometry, massive enhancements of the optical field can be achieved. As a result, measurable conversion from optical power to electric power can be obtained from a non-pulsed, low power (less than 1 mW) laser near 1550 nm. Optical rectification using AlGaAs waveguide has also been proposed for ultra-high-speed microwave-detection (Jinye Zhang, et al., *IEEE PHOTONICS TECHNOLOGY LETTERS*, VOL. 19, NO. 24, DECEMBER 15, 2007, 2012 - 2014, Microwave Photonic Signal Detection Using Phase-Matched Optical Rectification in an AlGaAs Waveguide). Using electrically or optically poled fibers is another way to enhance $\chi^{(2)}$ while maintaining high optical confinement and thus high intensity of light. Using a fiber coated with $\chi^{(2)}$ material is yet another way to achieve both large $\chi^{(2)}$ and high light intensity (Biqiang Jiang, et al., *Light: Science & Applications* volume 9, Article number: 63 (2020), High-efficiency second-order nonlinear processes in an optical microfiber assisted by few-layer GaSe). It has also been

proposed to use asymmetric coupled quantum wires for realizing obvious nonlinear optical rectification by breaking the inversion symmetry (Guanghui Liu, et al., Results in Physics, Vol. 17, June 2020, 103027, Nonlinear optical rectification and electronic structure in asymmetric coupled quantum wires).

To date, optical rectifiers have been proposed and experimentally demonstrated mainly for applications such as terahertz (THz) generation and high speed photodetection (e.g. US Patent No. 20090303574 A1, December 10, 2009. Peter Gunter, et al., Broadband Terahertz Radiation Generation and Detection System and Method). The device mentioned above uses the inorganic semiconductor ZnTe (zinc telluride) for the optical rectification and uses a femtosecond pulse laser to achieve THz radiation. However, remote charging using optical rectifier based on second order nonlinear materials has not been reported yet.

It is worth noting that the invented remote charging method is different from the reported methods of remote charging described above, such as those based on inductive coupling, RF energy harvesting, and surface plasmons. Based on the descriptions above, it is easy to understand that the preferred embodiment provides several advantages and overcomes several limitations of the existing methods of remote charging. First, the preferred embodiment can easily achieve remote charging over a long distance since a laser source is used. Laser beams can propagate in free space over a long distance with negligible spreading and loss if a proper optics and wavelength are used.

Second, the preferred embodiment can complete the charging process at the speed of light. As mentioned above, the voltage (of the optical rectifier) is proportional to the electric field (of the laser light coupled into the optical rectifier). Therefore, the electric field is generated instantly when the laser light reaches the optical rectification medium (i.e. independent of the electric circuit connected to the optical rectifier). From this, we can determine that voltage can be generated at the speed of light and is not affected by the load circuit connected to the optical rectifier. This feature of the preferred embodiment makes it different from other remote charging sources and batteries in its unique ability to charge extremely quickly, regardless of the attached load circuit.

Third, the preferred embodiment can easily maximize optical rectification by selecting the proper polarization/orientation and type of laser beam and nonlinear medium so that the largest $\chi^{(2)}$ component can be used.

Fourth, the preferred embodiment only needs to focus the light beam onto the $\chi^{(2)}$ nonlinear medium, without the need to couple the light asymmetrically on the two electrodes. This provides a major advantage over the surface plasmonic based optical rectification in which light must be coupled asymmetrically into the device.

Finally, the preferred embodiment does not need to couple the light into a waveguide structure with a nanoscale dimension, making packaging the device much easier. Therefore, similarly to the structures used in supercapacitor batteries, the gap between the two electrodes (i.e. the thickness of the nonlinear medium) can be very small, and a fold/spiral structure can be used to enhance the total surface area of the electrodes and thus the capacitance of the device.

As an example, the structure of the optical rectifier **1** is shown in **Fig.5** In the figure, the nonlinear material **2a** is a thin sheet, being sandwiched by two thin film electrodes **2b** to form an overall spiral structure. The nonlinear sheet **2a** is separated by an electrical isolator **2c**. The optical rectifier **1** is connected to a circuit through two metal wires **3** at the end of the spiral

structure. The light source, focus optics, and electric load are not shown in **Fig.5** since they are similar to those shown in **Fig.4**. The focus optics can be packaged together with the optical rectifier so that precise alignment between the laser source and optical rectifier is unnecessary. It is worth noting that, in this spiral structure, the nonlinear material is polarization insensitive, allowing the generation of the potential difference between the two sides of the nonlinear sheet material no matter the polarization direction of the light. In addition, the cross section of the spiral structure could be much larger than the wavelength of the incoming light, implying that efficient coupling of light into the spiral optical rectifier can be achieved easily.

A beam of light from a pulse laser is focused on the spiral optical rectifier by the focusing optics. The spiral optical rectifier **1** can generate a DC electric field within the nonlinear material sheet **2a** and create positive and negative charges on the two thin film electrodes **2b** of the nonlinear material sheet **2a**. Since the optical rectifier structure is a spiral, polarization insensitive $\chi^{(2)}$ material must be used. In fact, $\chi^{(2)}$ materials approaching polarization insensitivity have been reported in semiconductors such as GaAs (C. Kieleck, et al., Proc. SPIE (2010), 7582, Polarization effects and fiber-laser-pumping of a 2- μ m-pumped OP-GaAs OPO; P. S. Kuo, et al., OPTICS LETTERS, Vol. 32, No. 18, 2735 – 2737, GaAs optical parametric oscillator with circularly polarized and depolarized pump). The light source can remotely provide necessary energy to the optical rectifier through the focus optics and convert light energy to electric energy through the optical rectifier. The electric load receives electrical energy from the light source and uses the electrical energy to power the electric load (such as an IC circuit or re-chargeable battery).

Fig.6 demonstrates the simulation results of the voltage (V_{OR}) of the spiral optical rectifier as a function of the average power ($P_{Average}$) of a pulse laser. In the simulations, a width of 2 nm was assumed for both the nonlinear material sheet and insulation sheet. A nonlinear coefficient of 135 pm/V and a dielectric constant of 12.9 were used to imitate the material properties of GaAs (Torbjorn Skauli, et al., Optics letters. 27. 628-30. 10.1364/OL.27.000628, Measurement of the nonlinear coefficient of orientation-patterned GaAs and demonstration of highly efficient second-harmonic generation.). The diameter and height of the overall spiral structure were set as 2.2 cm and 10 cm, respectively. A femtosecond pulse laser with a repetition rate of 10 Hz and a pulse duration of 10 fs were assumed in the simulations. As shown in **Fig.6**, V_{OR} increases linearly with the increase of the average laser power. It is worth noting that a V_{OR} as high as 73.6 V can be achieved for 1000 W average laser power. This simulated voltage is higher than the standard voltage produced by the lithium-ion batteries on the market.

Fig.7 shows a simulated voltage (V_{Output}) across the spiral optical rectifier as a function of the discharging time (t). In the simulations, the same parameters as those used in **Fig.6** were employed, but the V_{OR} achieved at average power of 1000 W was assumed as the starting voltage of the discharging process. As shown in **Fig.7**, the voltage across the spiral optical rectifier decreases (at a decreasing rate) with the increase of time. It takes about 7.5 hours for V_o to drops down to 1/e of the initial voltage V_{OR} . Such a long discharge time is comparable to the standard discharging time of the lithium batteries on the market.

It is worth mentioning that the simulations shown in **Fig.6** and **Fig.7** are just some examples using GaAs nonlinear material and fs laser reported in previous literature with some assumptions on the spiral structure. However, it must also be noted that by employing nonlinear materials with higher nonlinear coefficients, lasers with higher average power and

optimized structures for the optical rectifier, higher V_{OR} and longer discharging time are possible.

In the second preferred embodiment, the structure of the device is demonstrated in **Fig.8**. As shown in **Fig.8**, multiple optical rectifiers are used. As an example, two optical rectifiers **1a** and **1b** are used and connected in parallel. The light source, integrated optics (for each optical rectifier), metal wires and electric load are not shown in **Fig.8** since they are similar to those shown in **Fig.4**. Similar to the case shown in **Fig.5**, polarization insensitive nonlinear material with a spiral structure can be used for the optical rectifiers. Differing from the first preferred embodiment, the second preferred embodiment demonstrated in **Fig.8** increases the total capacitance of the optical rectifiers since they are connected in parallel, resulting in a larger total capacitance. As a result, the discharge time is elongated through the relationship between the total capacitance and the discharge time. In addition, if the current generated from each optical rectifier remains the same, through the multiple optical rectifiers placed in parallel, the total current flowing through the electric load will increase, allowing for applications with a higher current requirement.

A collimated beam of light from a light source can cover all the optical rectifiers and the light will be further concentrated on each optical rectifier by focusing optics. This can be done by packaging a focus optics with each individual optical rectifier, i.e. each lens will focus light onto one optical rectifier to increase light intensity to generate a higher voltage (within the optical rectifier). The focusing optics can be packaged together with each optical rectifier so that precise alignment between the laser source and each spiral structure is unnecessary.

As described above, for each spiral optical rectifying structure, a laser beam from the pulse laser is focused upon the structure with a focusing optics. Each focusing optic (i.e. focusing lens) will converge the collimated incoming light beam to increase the optical intensity of the light beam and decrease the diameter of the light beam. As a result, with the focusing optics, the optical rectifying spiral structure will be able to generate a higher voltage and thus larger current. The focusing optics can be packaged together with each optical rectifier so that precise alignment between the laser source and each spiral structure is unnecessary. Although the collimated incoming laser beam should cover all the optical rectifiers in the second preferred embodiment, since the incoming beam size is usually much bigger than the overall size of optical rectifiers after propagating a long distance, it is reasonable to assume that each beam of light focused on each optical rectifier in the second preferred embodiment have the same optical intensity as that in the first preferred embodiment.

The optical rectifiers **1a** and **1b** can generate DC electric fields within each structure and will create positive and negative charges on the two end surfaces of the spiral structure, respectively. Since the optical rectifiers **1a** and **1b** are connected in parallel, and thus can result in a larger capacitance and a larger current as compared to the case shown in **Fig.4**. When the optical rectifiers **1a** and **1b** are connected to an electric load through the metal wires **3**, electrons will be driven to flow through the electric load. This effect will result an electric current (DC current) following through the load. In other words, the light source can remotely provide the necessary energy to the optical rectifiers through the focus optics and convert light energy into electric energy through the optical rectifiers. The electric load receives electrical energy from the light source and uses the electrical energy to power the electric load (such as a re-chargeable battery).

As described above, the optical rectifier generates a DC electric field within the spiral structure (as the light beam is focused on the structure), resulting positive and negative

charges separated on the two opposite electrodes of spiral. As shown in **Fig.5**, by enlarging the spiral's "tail," enough space will be generated so that the anode and cathode electrode can be attach easily to the circuit through the conducting wires **3** and allow electrons to flow through the electric load, i.e. producing a DC current (through the load). As shown in **Fig.8**, the optical rectifiers **1a** and **1b** are positioned in parallel. Thus, the DC current generated from **1a** can be added to the DC current generated from **1b** to result in a larger DC current, assuming the voltage generated in each optical rectifier is the same. In addition, since the total capacitance for a circuit (in parallel) is the summation of parallelly connected capacitor, the capacitance for **1a** and **1b** can be added together, resulting in a larger total capacitance. Considering the fact that the greater the capacitance, the longer the discharge time, the second preferred embodiment will have a longer discharge time as compared to the first preferred embodiment.

Similar to the first preferred embodiment, the second preferred embodiment includes many different optimizable parameters. These variables include the width of nonlinear medium and isolator, height and diameter of spiral, length of spiral is not included as it can be determined through the diameter and width of the spiral. These variables greatly influence the voltage (within the optical rectifier) and the discharge time. For example, as the height has no influence on the generated voltage and will increase the discharge time when it increases. Therefore, for applications requiring a long discharge time and can allow for a large power source, increasing the value of the height would be greatly beneficial. Since all of the variables listed above influence at least one or both voltage and discharge time, the second preferred embodiment has great potential to be optimized for different applications.

From the above descriptions, it is easy to understand that the second preferred embodiment provides the same advantages and overcomes the same limitations of the existing methods of energy storage and remote charging as the first preferred embodiment. These advantages include a large capacitance (which results in a longer discharge time), a structure which is polarization independent, and extremely optimizable features. The second preferred embodiment of the present invention requires several parallelly connected optical rectifiers (with a spiral structure) and a light source with multiple focus optics (for each optical rectification structure) to achieve remote charging. Since the $\chi^{(2)}$ material used for optical rectifiers is polarization insensitive to the polarization direction of the light source, a laser with randomly polarized light can be used. In addition, the second preferred embodiment provides a larger capacitance and a larger energy storage capacity (or larger remote charging capabilities). This implies that the second preferred embodiment of the present invention can provide a larger current to the load and longer discharging time as compared to the first preferred embodiment.

The above embodiments have described remote charging and energy storage methods based on laser illumination and optical rectification. Of course, the methods described in the present invention can be applied to other applications. For example, the methods described in the present invention can be applied to rechargeable batteries. The methods are not limited to remote charging and energy storage. The layout and configuration of the components involved in the present invention can also be altered to fit different shapes and to perform different functions.

6. CLAIMS

What is claimed is:

1. A remote charging system which consists of a light source, a packaged focus optics, an optical rectification device, and an energy receiver to achieve functions of remote charging.
2. The light source in **Claim 1** is a coherent light source with a polarization direction and a wavelength, matching with the best performance of the optical rectifier in **Claim 1**, the power of the coherent light source ranges from 1 W to 1 kW.
3. The light source in **Claim 1** is a pulse laser with a pulse duration of femtosecond and with a wavelength in a range of UV, visible, and mid-infrared.
4. The light source in **Claim 1** is linearly polarized light or randomly polarized light.
5. The packaged focus optics in **Claim 1** are focal lenses to focus the light beam from the light source in **Claim 2** into a small circular spot to match the shape and size of the optical rectification device in **Claim 1**.
6. The optical rectification device in **Claim 1** is single optical rectifier formed on a nonlinear material with a high second-order nonlinearity, which is sandwiched between two electric conductive electrodes.
7. The optical rectification device in **Claim 1** is formed by multiple optical rectifiers which are arranged in parallel, and each optical rectifier is formed on a nonlinear material with a high second-order nonlinearity and has a packaged focus optics.
8. The optical rectification device in **Claim 1** is made from nanowires for efficient optical rectification.
9. The optical rectification device in **Claim 1** is wavelength dependent and can only rectify the light in the UV to the mid-infrared range.
10. The optical rectification device in **Claim 1** is wavelength insensitive and rectify the light source with a broadband wavelength range.
11. The optical rectification device in **Claim 1** is polarization dependent and requires a linearly polarized light source.
12. The optical rectification device in **Claim 1** is polarization independent and can rectify randomly polarized light.
13. The optical rectifier in **Claims 7 and 8** has an Archimedean spiral structure.
14. The energy receiver in **Claim 1** is an integrated circuit that needs electrical energy to operate.
15. The energy receiver in **Claim 1** is a rechargeable battery.